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Theory(Logic)

This program in C++ shows how to use this pointer. Create a class named `Coordinate` let its private Members be `latitude` and `longitude` to store the coordinates. Inside the class create a Constructor which takes two values: `latitude` and `longitude`. If no values are passed, both default to 0. `setLatitude()` and `setLongitude()`: These functions set the values of `latitude` and `longitude`. They return `*this`, which means they return the current object. In the main function: An object `location` is created with initial values (15, 30). Then, `location.setLatitude(40).setLongitude(70);` .Finally, `display()` prints the updated coordinates.

Pseudo Code

- **Step 1: Include input-output library**
- **Step 2: Use standard namespace**
- **Step 3: Create class Coordinate**
 - **Private:**
 - Integer latitude
 - Integer longitude
 - **Public:**
 - Constructor with default values (lat=0, lon=0)
 - Set latitude = lat
 - Set longitude = lon
 - Function setLatitude(lat)
 - Set latitude = lat
 - Return reference to current object
 - Function setLongitude(lon)
 - Set longitude = lon
 - Return reference to current object
 - Function display()
 - Print latitude and longitude
- **Step 4: Start main function**
 - Create object location with latitude=15 and longitude=30
 - Call setLatitude(40) and setLongitude(70) using chaining on location
 - Call display() to print the values
- **Step 5: End main function**

Program Code

```
#include<iostream>
using namespace std;
class Coordinate
{
    private:
        int latitude;
        int longitude;

    public:
        Coordinate(int lat=0,int lon=0)
        {
            this->latitude=lat;
            this->longitude=lon;
        }

        Coordinate& setLatitude(int lat)
        {
            latitude=lat;
            return *this;
        }

        Coordinate& setLongitude(int lon)
        {
            longitude=lon;
            return *this;
        }

        void display() const
        {
            cout<<"latitude= "<<latitude<<"\nlongitude= "<<longitude<<endl;
        }
};
```

```
int main()  
{  
    Coordinate location(15,30);  
    location.setLatitude(40).setLongitude(70);  
    location.display();  
    return 0;  
}
```

Output

```
latitude= 40  
longitude= 70
```

Conclusion

Code is run successfully, required output is received.